

EXPERIMENT 5

AIM:

To collect, load, and perform initial exploration of this Iris dataset using Python and data visualization libraries such as Pandas, Matplotlib and Seaborn.

CODE:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


from google.colab import files

uploaded = files.upload()


df = pd.read_csv("iris_dataset.csv")


print("First five rows of the dataset:")

print(df.head())


print("\nDataset Information:")

print(df.info())


print("\nSummary Statistics:")

print(df.describe())


df.hist(bins=20, figsize=(12, 8))
```

```
plt.tight_layout()
```

```
plt.show()
```

```
sns.pairplot(df, hue="Species")
```

```
plt.show()
```

OUTPUT:

First five rows of the dataset:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	SepalLengthCm	150 non-null	float64

```
1 SepalWidthCm 150 non-null float64
2 PetalLengthCm 150 non-null float64
3 PetalWidthCm 150 non-null float64
4 Species      150 non-null object
```

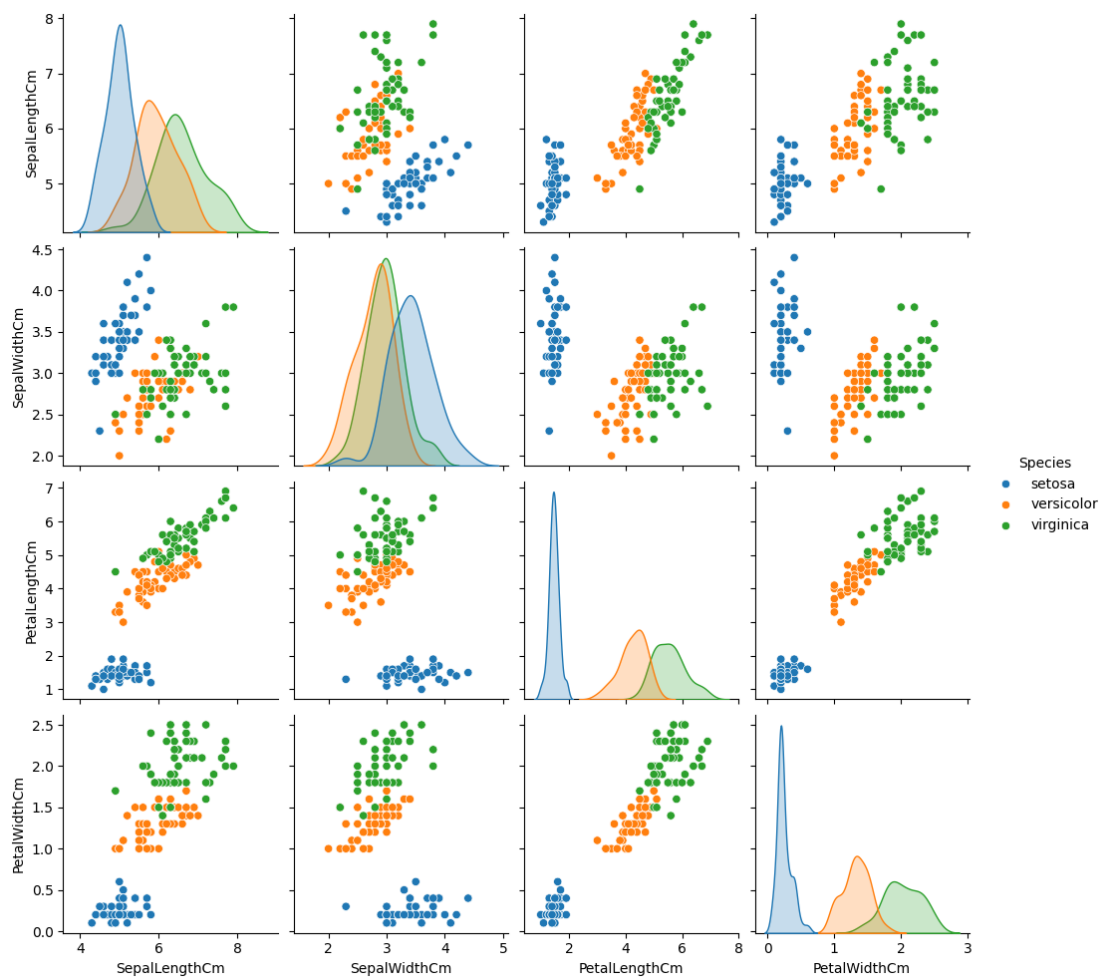
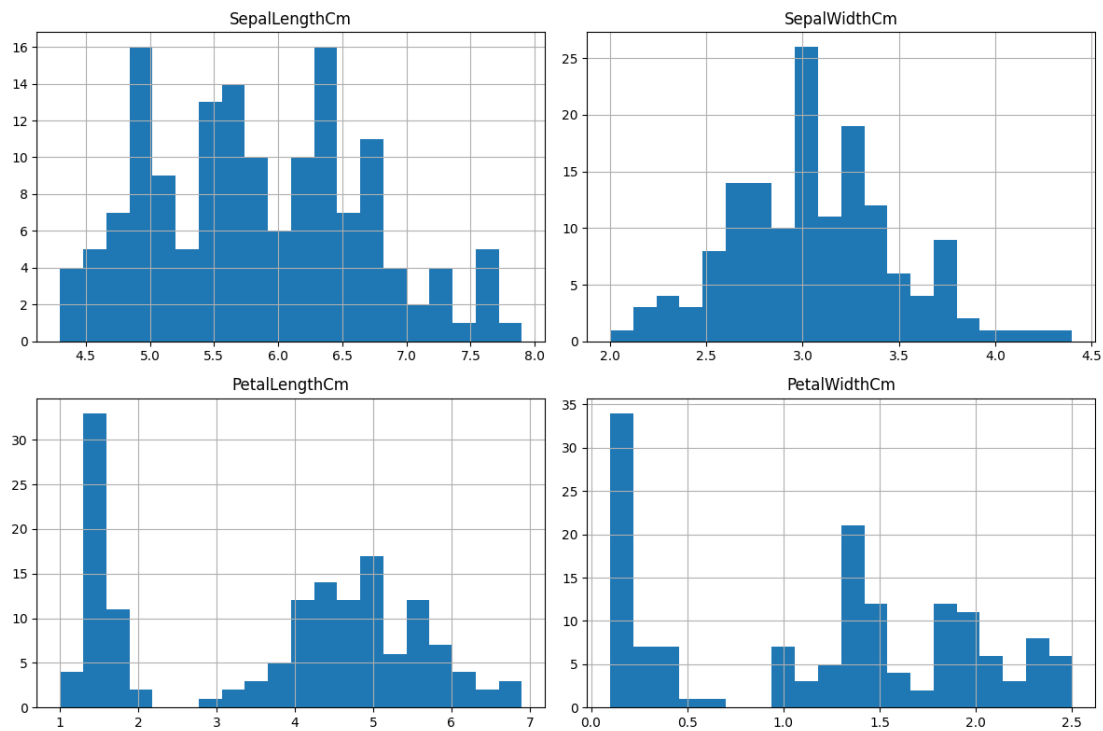
```
dtypes: float64(4), object(1)
```

```
memory usage: 6.0+ KB
```

```
None
```

Summary Statistics:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000



RESULT:

The Iris dataset was successfully loaded and explored. The dataset structure, summary statistics, feature distributions, and relationships between features were visualized using Pandas, Matplotlib, and Seaborn.